

A Novel Design for Power Reduction in Arithmetic Circuits using MTCMOS Technology

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Abstract

In CMOS logic circuits, the reduction in the threshold voltage due to voltage scaling leads to increase in the subthreshold leakage current and hence static power dissipation. Although power consumption is important for modern VLSI design, operation speed and occupied area are still the main requirements of the VLSI design. Multi threshold voltage CMOS (MTCMOS) technology is a good solution which provides a high performance and low-power design without any area overhead. MTCMOS technology provides the transistors that have low, normal and high threshold voltage. The low-threshold voltage transistors are used to reduce the propagation delay time in the critical path, the high-threshold voltage transistors are used to reduce the power consumption in the shortest path. This paper describes a low-power and high speed design for full adder, full subtractor and 4×4 carry save multiplier circuits with Multi-threshold CMOS (MTCMOS) technology using Microwind EDA tool.

Keywords: MTCMOS, full adder, full subtractor, CMOS circuit, leakage current, low power design.

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INTRODUCTION

The rapid increase in the number of transistors on chips has enabled a dramatic increase in the performance of computing systems. However, the performance improvement has been accompanied by an increase in power dissipation; thus, requiring more expensive packaging and cooling technology. Power dissipation in CMOS circuits has been the charging and discharging of load capacitances, often referred to as the dynamic power dissipation. Dynamic power is consumed when transistors are switching. As the technology continues to scale down a significant portion of the total power consumption in high performance digital circuits is due to leakage current because of reduced threshold voltage. Metal-oxide-semiconductor field-effect transistor (MOSFET)s are fabricated with high overall doping concentration, lowered source/drain junction depths, halo doping, high-mobility channel materials, etc. Furthermore, the reduction of the gate oxide thickness (t_{ox}) causes a drastic increase in the gate tunneling leakage current due to carriers tunneling through the gate oxide, which is

strong exponential function of the voltage magnitude across the gate oxide [1, 2] consequently, to minimize the leakage power in active mode.

In current CMOS technologies, the sub threshold leakage current is much larger than the other leakage current components. This current can be calculated by using the following equation:

$$I_{DS} = K (1 - e^{-V_{DS}/V_T}) e^{(V_{GS} - V_T + V_{DS})/V_T}$$

where, K is the drain-induced barrier lowering coefficient. Even in current-generation technology, sub threshold leakage power dissipation is comparable to the dynamic power dissipation, and the fraction of the leakage power will increase significantly in the near future. Today's microprocessor designs devote a large fraction of the chip area [3] to the memory structures. High-performance onchip caches are a crucial component in the memory hierarchy of modern computing systems. In this technique each NMOS and PMOS transistors in the logic gates is split into two transistors. Leakage

current flowing through the NMOS transistor stack reduces due to the increase in the source to substrate voltage in the top NMOS transistor and also due to an increase in the drain to source voltage in the bottom NMOS transistor. This reduces the power dissipation in logic circuits. This technique is implemented to BASIC gates such as AND, OR, XOR etc, COMBINATIONAL circuits such as FULL ADDER, SEQUENTIAL circuits such as D-Flip-flop and also for memory cells such as 6T1RAM CELL.

CIRCUIT DESIGN WITH MTCMOS TECHNOLOGY

Multi threshold voltage CMOS (MTCMOS) technology provides a solution to the high performance and low power design requirements of modern designs. MTCMOS technology provides the transistors that have low, normal and high threshold voltage [4]. This technology is an effective circuit level technique that provides a high performance and low-power design by utilizing both low and high threshold voltage transistors. Low-threshold voltage transistors have high-speed

performance but high-power consumption. High-threshold voltage transistors have low-power consumption but low-speed performance. While the low-threshold voltage transistors are used to reduce the propagation delay time in the critical path, the high-threshold voltage transistors are used to reduce the power consumption in the shortest path [5–7]. This paper describes a low-power and high speed design for full adder, 4-bit ripple carry adder and 4×4 multiplier circuits with MTCMOS technology.

Design of Full Adder

When we add three bits A, B and Cin (input carry), then the Boolean function of the sum and carry are given as following:

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{Carry} = AB + (A \oplus B) C$$

Figure 1 shows the logic diagram of 1-bit full adder cell. The full adder contains 3 inputs (A, B, Cin) and 2 outputs (Sum and Carry). For the full adder circuit, carry path is the longest path and sum path is the shortest path.

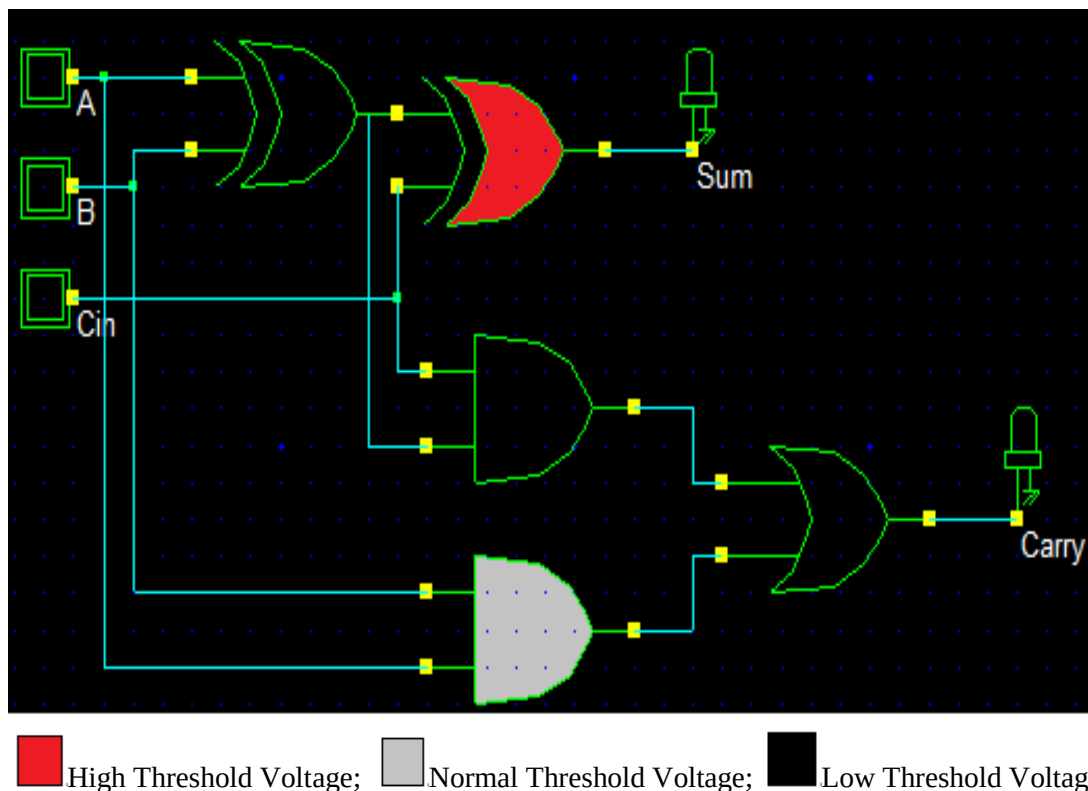


Fig. 1: Logic Diagram for 1-bit Full Adder.

We use low-threshold voltage transistors, normal-threshold voltage transistors and high-threshold voltage transistors in the circuit design. Since carry path is the longest path in the circuit, the low-threshold voltage transistors are used in this path to reduce the propagation delay time in the critical path. The second exclusive-OR gate present in the sum path is designed with high-threshold voltage transistors to reduce the power consumption in the shortest path. The remaining AND gate is

designed with normal-threshold voltage transistor.

Design of Full Subtractor

A full subtractor is a combinational circuit that performs a subtraction between two bits taking into account that a 1 may have been borrowed by a lower significant stage. This circuit has three inputs and two outputs. The three inputs A, B and C denote the minuend, subtrahend and previous borrow, respectively. Figure 2 shows logic diagram for 1-bit full subtractor.

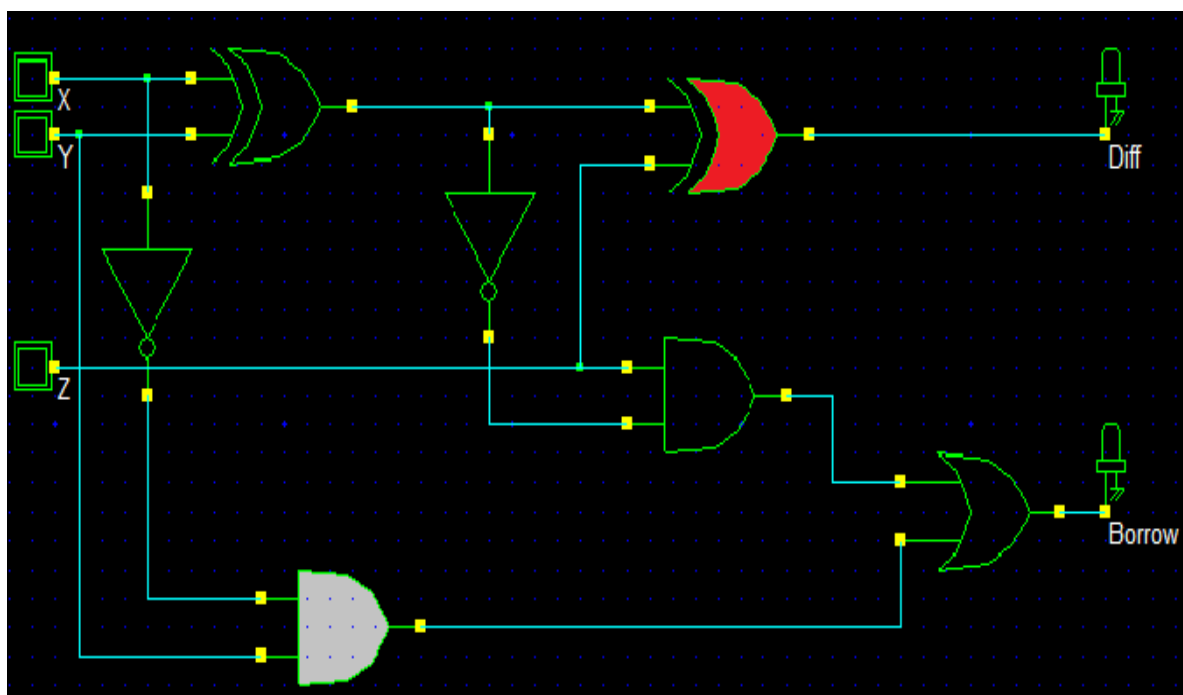


Fig. 2: Logic Diagram for 1-bit Full Subtractor.

The two outputs D and BORROW represent the difference and borrow, respectively. The logic circuit for full subtractor is shown in Figure 2.

Design of 4-bit Carry Save Multiplier

The 4×4 multiplier circuit 8 inputs (a0, a1, a2, a3, b0, b1, b2 and b3) and 8 outputs (P0, P1, P2, P3, P4, P5, P6 and P7). The output P7 is the critical path (longest path) in this circuit. This critical path contains two half adders and four full adders as shown in the figure. The carry paths in each of these six adders are designed with low-threshold voltage transistors to reduce the propagation delay time in the critical path.

The sum path in each of these six adders (the exclusive-OR gates in each of these six adders which gives the outputs P1, P2, P3, P4, P5 and P6) and the AND gate which gives the output P0 are designed with high-threshold voltage transistors to reduce the power consumption in the shortest paths. The remaining gates are designed with normal-threshold voltage transistors.

PROPOSED DESIGN WITH MTCMOS TECHNOLOGY

Full Adder

The full adder design is proposed by eight transistors only is shown in Figure 3.

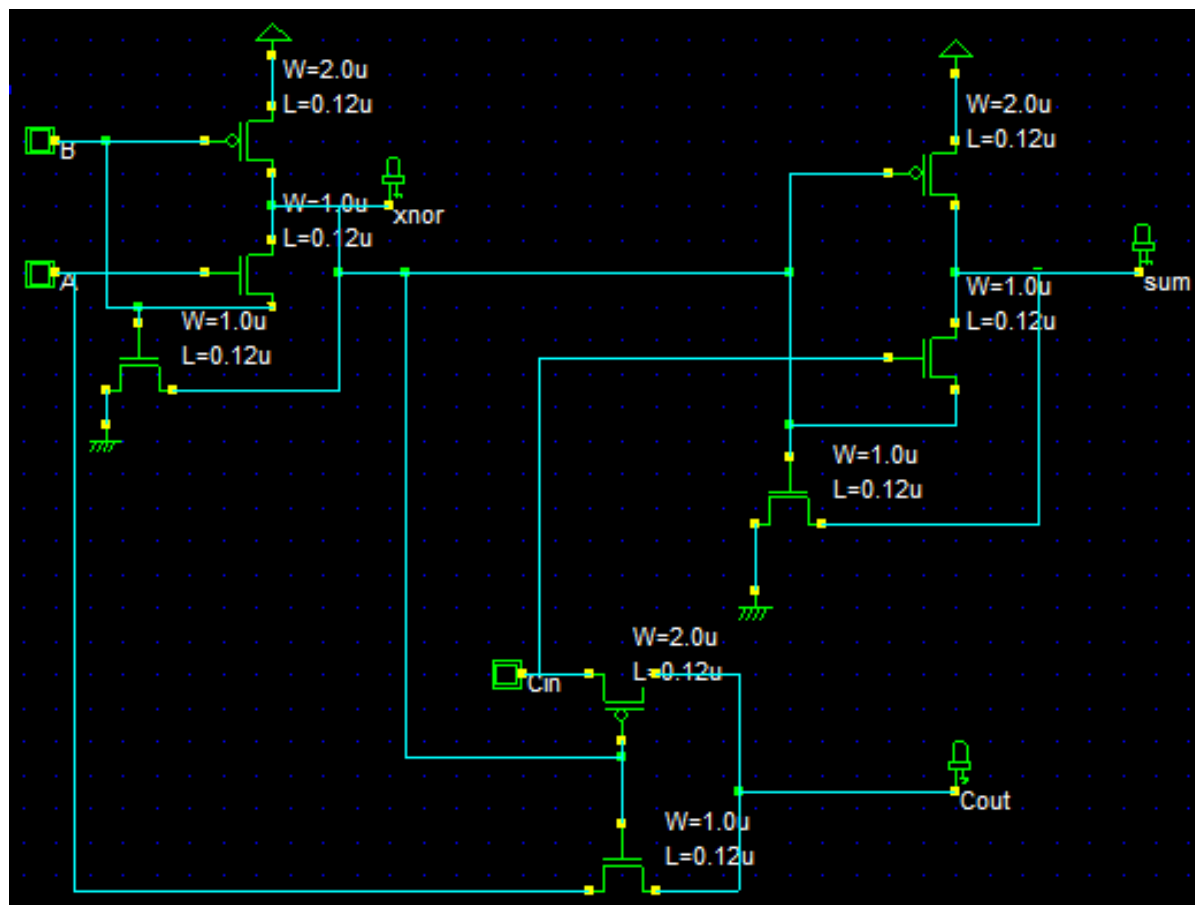


Fig. 3: Proposed Design of Full Adder.

The output waveform for full adder is shown in Figure 4.

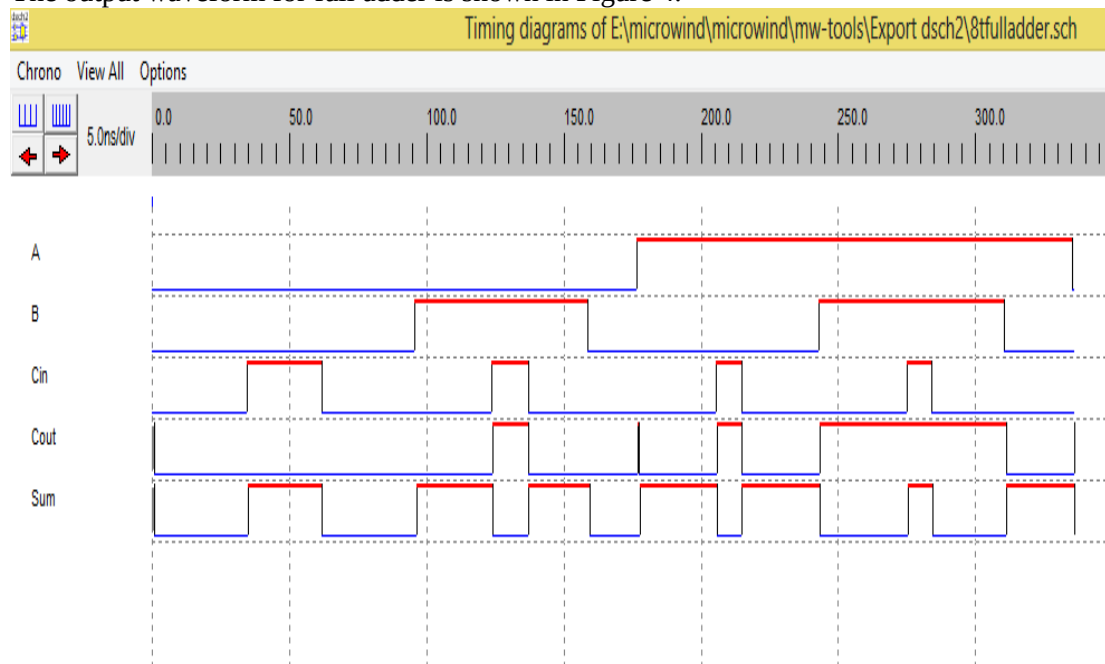


Fig. 4: Waveform for Full Adder.

Full Subtractor

The full subtractor as shown in Figure 5 is proposed by only 14 transistors only.

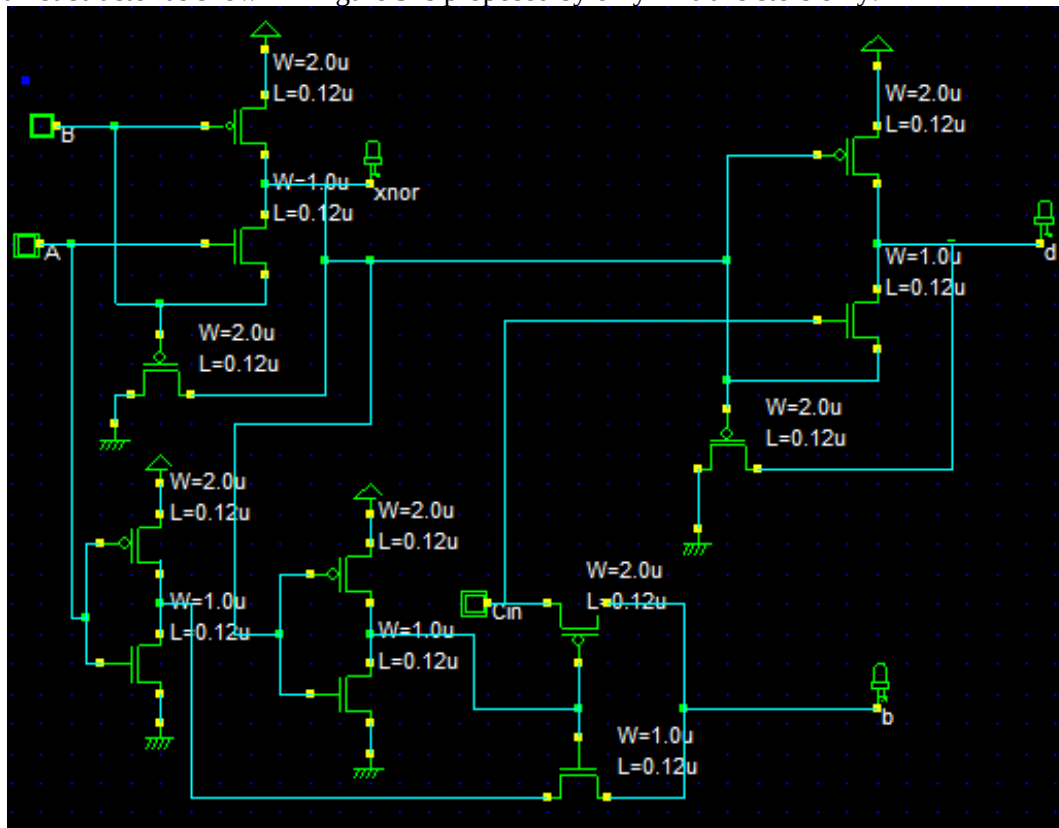


Fig. 5: Proposed Design of Full Subtractor.

The waveform for full subtractor is shown in Figure 6.

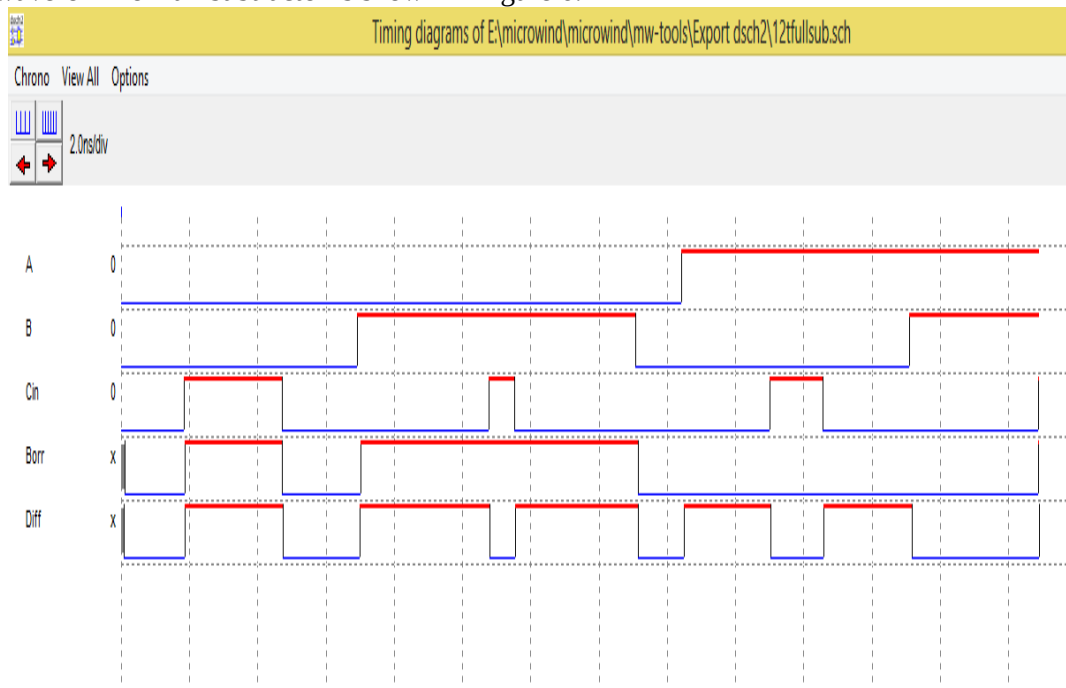


Fig. 6: Wave Form for Full Subtractor.

4*4 Carry Save Multiplier

The proposed 4*4 carry save multiplier is shown in Figure 7.

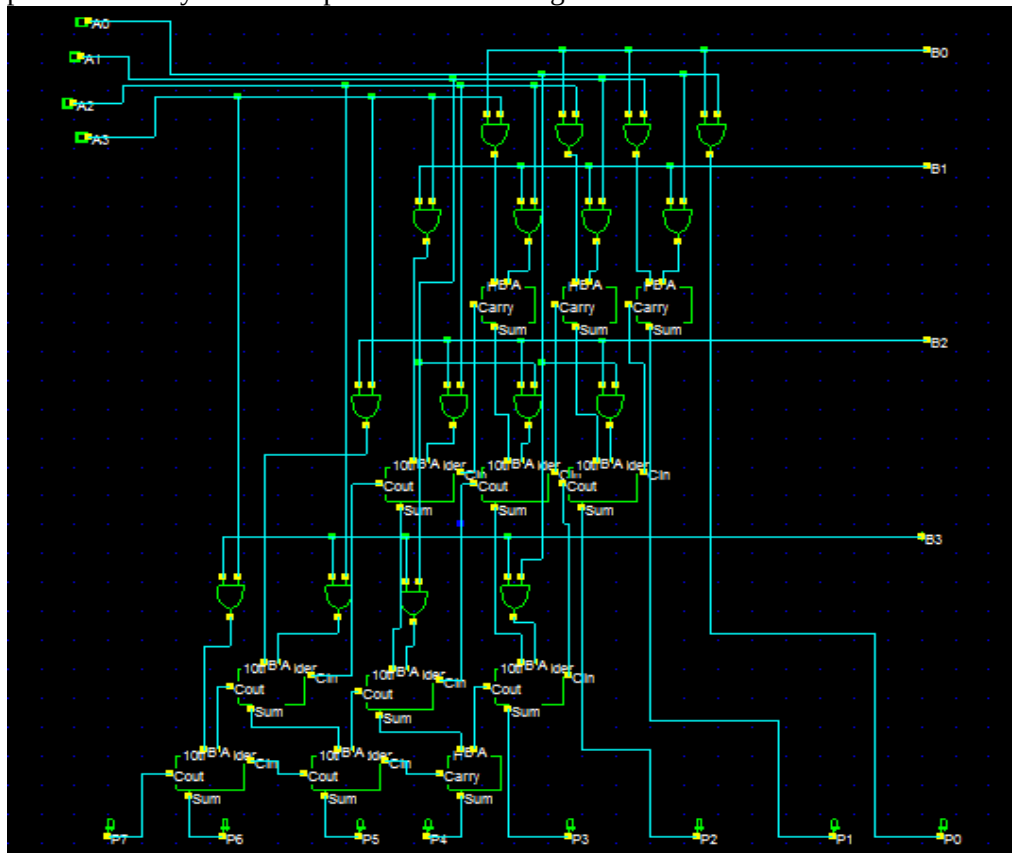


Fig. 7: Proposed Design of Carry Save Multiplier.

The waveform for 4*4 carry save multiplier is shown in Figure 8.

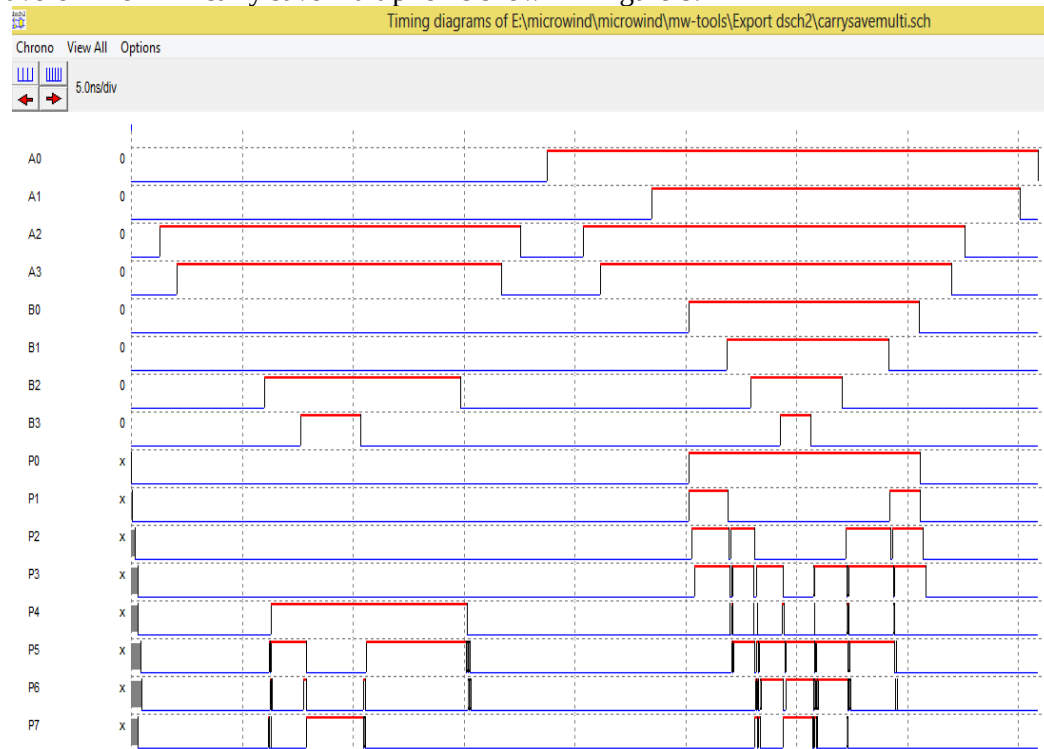


Fig. 8: Wave form for 4*4 Carry Save Multiplier.

RESULTS

The proposed full adder, ripple carry adder and multiplier circuits are simulated through Microwind EDA tool. The supply voltage for 45 nm technology is 1.0V [1].

Table 1 shows the comparison table between the conventional full adder circuit and the proposed circuit. The comparison between the conventional 4-bit ripple carry adder and the proposed circuit is shown in the Tables 2 and 3 show the comparison table between the conventional 4×4 multiplier circuit and the proposed circuit.

Table 1: Comparison Table for Full Adder Circuit.

Parameter	Conventional circuit	MT CMOS
Static power (watts)	87.792 E-6	0.101 E-6
Area (um ²)	7254	6090

Table 2: Comparison Table for Full Subtractor Circuit.

Parameter	Conventional circuit	MT CMOS
Static power (watts)	83.195 E-6	0.875 E-6
Area (um ²)	8778	7250

Table 3: Table For 4*4 Carry Save Multiplier Circuit.

Parameter	MTCMOS
Static power (watts)	0.243 E-3
Area (um ²)	272490

CONCLUSIONS

In this paper, low-power and high speed CMOS logic circuits are designed using MTCMOS technology. The propagation delay time in the critical path is reduced by using low-threshold voltage transistors.

The power consumption is reduced in the shortest path by using high-threshold voltage transistors. Since there is no additional

circuitry added in this technique, there will be no area overhead to the proposed circuits.

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